

Coupon Collecting Problem

n different "coupons"

each day get 1, randomly chosen from the n

$X = \#$ of days ^{to} ~~before~~ get all n

$$X = X_1 + X_2 + \dots + X_n$$

$$X_1 \sim \text{Geometric}(p = 1)$$

$$X_2 \sim \text{Geometric}(p = \frac{n-1}{n})$$

$\vdots$

$$X_i \sim \text{Geometric}(p = \frac{n-i+1}{n})$$

Don't say "95% CI for X "

Instead "95% prediction interval for X "

i.e. L, U such that $P(L \leq X \leq U) = 0.95$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Exponential(λ)
 2 distribution. Derive the MLE for λ .

$$f_X(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

$$L(\lambda) = \prod_{i=1}^n f_{X_i}(x_i; \lambda)$$

$$= \prod_{i=1}^n \lambda e^{-\lambda x_i}, \quad x_i > 0$$

$$= \lambda^n e^{-\lambda \sum x_i}, \quad x_i > 0$$

$$\log L(\lambda) = n \log \lambda - \lambda \sum x_i$$

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum x_i$$

set equal to 0, $\hat{\lambda}_{MLE} = 1/\bar{x}$

$$\begin{aligned} \log a^b &= b \log a \\ \log ab &= \log a + \log b \\ \log e^a &= a \end{aligned}$$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Beta($\alpha, 1$) distri-
 2 bution. Derive the MLE for α . Compare this with the MOM estimator.

$$f_X(x; \alpha) = \alpha x^{\alpha-1}, \quad 0 < x < 1$$

$$L(\alpha) = \prod_{i=1}^n f_{X_i}(x_i; \alpha) = \alpha^n (\prod x_i)^{\alpha-1}, \quad 0 < x_i < 1$$

$$\log L(\alpha) = n \log \alpha + (\alpha-1) \log(\prod x_i)$$

$$\frac{\partial \log L(\alpha)}{\partial \alpha} = \frac{n}{\alpha} + \log(\prod x_i)$$

set equal to 0, find $\hat{\alpha}_{MLE} = -\frac{n}{\log(\prod x_i)} = -\frac{n}{\sum \log x_i}$

$$E(X) = \frac{\alpha}{\alpha+1}, \quad m_1 = \bar{X} \quad \hat{\alpha}_{mom} = \frac{\bar{X}}{1-\bar{X}}$$

- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. with the $\text{Normal}(\mu, \sigma^2)$ dis-
 2 tribution. Derive the MLE for μ and σ^2 .

$$3 \quad f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad -\infty < x < \infty$$

$$4 \quad L((\mu, \sigma^2)) = \prod_{i=1}^n f_{X_i}(x_i)$$

$$5 \quad = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$6 \quad = \left(\frac{1}{(2\pi)^{n/2}}\right) \left(\frac{1}{\sigma^2}\right)^{n/2} \exp\left(-\frac{\sum (x_i-\mu)^2}{2\sigma^2}\right)$$

$$7 \quad \ell((\mu, \sigma^2)) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{\sum (x_i-\mu)^2}{2\sigma^2}$$

$$1 \quad \frac{\partial \ell(\mu, \sigma^2)}{\partial \mu} = + \frac{2 \sum (x_i - \mu)}{2\sigma^2}$$

$$2 \quad \text{Set equal to 0, } \hat{\mu}_{MLE} = \bar{X}$$

$$3 \quad \frac{\partial \ell(\mu, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum (x_i - \mu)^2}{2\sigma^4}$$

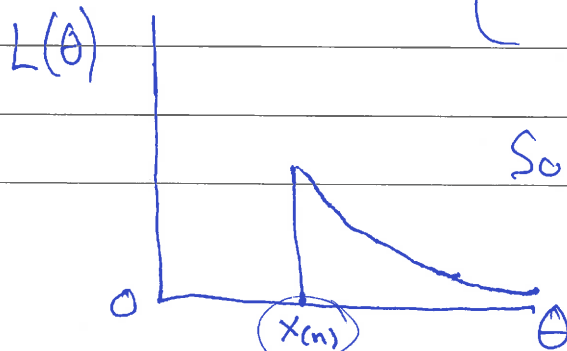
$$4 \quad \text{Set equal to 0, } \hat{\sigma}_{MLE}^2 = \sum (x_i - \bar{X})^2 / n \neq s^2$$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Uniform}(0, \theta)$. Derive the
 2 MLE for θ . What is the MOM in this case? "German Tank Problem"

$$f_X(x) = \left(\frac{1}{\theta}\right) \mathbf{1}_{\{0 \leq x \leq \theta\}}$$

$$L(\theta) = \prod_{i=1}^n f_{X_i}(x_i) = \prod_{i=1}^n \left(\frac{1}{\theta}\right) \mathbf{1}_{\{0 \leq x_i \leq \theta\}}$$

$$= \begin{cases} \left(\frac{1}{\theta}\right)^n, & 0 \leq x_i \leq \theta \text{ for all } i \\ 0, & \text{otherwise} \end{cases}$$



So, $\hat{\theta}_{MLE} = X_{(n)}$ (the max. of the X 's)

1 Note that $\hat{\theta}_{MLE} \leq \theta$, which is not appealing

$$2 \text{ mom: } E(X) = \frac{\theta}{2} \quad m_1 = \bar{X}$$

$$3 \hat{\theta}_{mom} = 2\bar{X}$$

6 Problem: Nothing preventing $2\bar{X} < X_{(n)}$

$$8 \text{ Could use } \hat{\theta}_3 = \max(2\bar{X}, X_{(n)})$$

$$10 \text{ Or: Can show } E(X_{(n)}) = \left(\frac{n+1}{n+2}\right)\theta$$

$$11 \hat{\theta}_4 = \left(\frac{n+2}{n+1}\right)(X_{(n)})$$

1 Finding the MLE Numerically

2 In some situations it is not possible to derive the MLE using an analytical
3 approach. Instead, we can use a numerical approach to optimizing the
4 likelihood.

5 The primary limitation with this approach is that one does not find a gen-
6 eral form for the estimator. Instead, the result is the estimate for the one
7 data set under consideration.

8 When taking a numerical approach, it is still a good idea to work with
9 the log likelihood, as this function is usually better behaved. (The product
10 in the likelihood function will often lead to a likelihood with either very
11 small or very large values.)

1 R includes general-purpose optimization capability; here we will focus on
2 `optim()`, which minimizes a function specified by the user.

3 The trickiest thing about using `optim()` is to ensure that the function to
4 be minimized is supplied in the appropriate format.

5 We'll begin with a simple toy example. Suppose that X is $\text{binomial}(n, p)$,
6 and that p is unknown. We've seen that the sample proportion $\hat{p} = X/n$
7 is unbiased and has standard error of $\sqrt{p(1-p)/n}$. What value of p maxi-
8 mizes this standard error? Of course, we know that $0 \leq p \leq 1$.

9 Let's start with the step of building the function which is to be minimized.

```
> negseofphat = function(p, n) Optimizing over p
+ {
+   return((-1) * sqrt(p * (1-p) / n))
+ }
```

1 Comments on this:

- 2 1. `optim()` **minimizes** functions, so I have this function return -1 times
3 the standard error. Minimizing this function will correspond to maxi-
4 mizing the standard error.
- 5 2. It is crucial that the first input argument to the function to be op-
6 timized (in this example, `negseofphat()`) be the variable(s) over
7 which the function is to be optimized. If there is more than one such
8 variable, then the variables must come in as a vector. We'll do such an
9 example below. In this example, we are minimizing over p , so `p` goes
10 in as the first argument.
- 11 3. There can be any number of additional arguments to the function, but
12 these must start with the second in the list. These can have default
13 values, or their values will get passed in from `optim()`.

1 Now, we'll actually call `optim()` to do the optimization:

```
> holdopt = optim(0.1, negseofphat, n=10,  
+               method="BFGS", hessian=T)  
>  
> print(holdopt$par)  
[1] 0.500004
```

2 Comments on this:

- 3 1. The first argument to this function (0.1 in this example) is the starting
4 values for the parameter(s) over which the function will be minimized.
5 This should be on the interior (not the boundary) of the space of pos-
6 sible values for the parameter(s). If there is more than one parameter,
7 then this argument should be a vector.

- 1 2. The second argument is the function to be minimized. Note that there
2 are no parentheses; it's simply the name of the function.
- 3 3. Any arguments which are to be passed on to the function to be mini-
4 mized are then listed. Use the same name as the argument has in that
5 function. For example, the second argument to `negseofphat()` is `n`,
6 in the example call I specify `n = 10`.
- 7 4. The method argument tells `optim()` which minimization technique
8 to utilize. The default is the Nelder-Mead method, but this does
9 not work well with one-dimensional minimization problems. See
10 `help(optim)` for more details.
- 11 5. I am requesting that the Hessian of the function (evaluated at the min-
12 imum) be returned by the algorithm. We will see in a future lecture
13 why this is important.

1 Numerical Maximization of Likelihood in Gamma Case

2 Here we are assuming that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Gamma}(\alpha, \beta)$. The
3 following function will return the MLE when a vector of observations are
4 passed in. The Hessian evaluated at the maximum is also returned.

5 This function can be loaded using

```
> source("http://www.stat.cmu.edu/~cschafer/MSCF/gammamle.txt")
```

6 The code is shown on the following two slides. Note how the MOM esti-
7 mator is used as a starting point for the optimization algorithm.

```

> # The function to find the MLE of the pair of parameters
> # in the gamma distribution. The input x is a vector of
> # observations assumed to be realizations of iid Gamma
> # random variables.
>
> gammamle = function(x)
+ {
+
+ # The negative of the log-likelihood. This is the function
+ # that will be minimized via optim().
+
+   negloglikelihood = function(pars, x)
+   {
+     return((-1) * sum(dgamma(x, pars[1], pars[2], log=T)))
+   }
+
+   
$$\sum \log f_{x_i}(x_i)$$

+
+ # Set the starting values for the algorithm at the method

```

Handwritten notes on Slide 25:
 Blue arrows point from the parameters (α, β) to `pars[1]` and `pars[2]` in the `dgamma` function call.
 A blue bracket on the right side of the `return` statement groups the calculation of the negative log-likelihood.
 The formula $\sum \log f_{x_i}(x_i)$ is written below the `return` statement, with an arrow pointing to the `sum` function.

```

+ # of moments estimates.
+
+   betahatmom = mean(x) / (mean(x^2) - mean(x)^2)
+   alphahatmom = betahatmom * mean(x)
+
+ # The optimization.
+
+   optimout = optim(c(alphahatmom, betahatmom),
+                     negloglikelihood, x=x, hessian=T)
+
+   return(list(mle = optimout$par, hessian=optimout$hessian))
+ }

```


- 1 **Exercise:** Test `gammamle()` by simulating some data sets where the true
 2 values of the parameters are known.

3 Examples: $n=100$, $\alpha=2$, $\beta=4$, $\hat{\alpha}=2.036$, $\hat{\beta}=3.708$

4 $n=1000000$, $\hat{\alpha}=2.0015$, $\hat{\beta}=4.0009$

10

1 **Example: Regression with t-distributed Errors**

- 2 In this example, we will consider the following regression model: The re-
 3 sponse Y is related to the single predictor by

4
$$Y = \beta_0 + \beta_1 x + \sigma \epsilon$$

$$T \sim t_\nu$$

$$V(T) = \frac{\nu}{\nu-2}$$

- 5 where ϵ is assumed to have the t -distribution with ν degrees of freedom.
 6 We will assume that ν is set by the user (not estimated).

- 7 There are three unknown parameters in this model: β_0 , β_1 , and σ . We seek
 8 to estimate these via maximum likelihood.

- 9 Note that σ is not exactly equal to the standard deviation of the irreducible
 10 errors since the t -distribution does not have variance of one. Nevertheless,
 11 σ is closely related to the standard deviation of the irreducible errors.

- 1 **Exercise:** We assume that we will observe n pairs of (x, Y) values, and that
 2 the ϵ values are independent. Derive the density for a single Y observation.

$$Y = \beta_0 + \beta_1 x + \sigma \epsilon$$

$$\begin{aligned} P(Y \leq y) &= P(\beta_0 + \beta_1 x + \sigma \epsilon \leq y) \\ &= P\left(\epsilon \leq \frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right) \\ &= F_T\left(\frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right), \text{ where } T \sim t_\nu \end{aligned}$$

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = f_T\left(\frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right) \left(\frac{1}{\sigma}\right)$$

$$L((\beta_0, \beta_1, \sigma)) = \prod_{i=1}^n f_T\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right) \left(\frac{1}{\sigma}\right)$$

- 1 In preparation for using `optim()`, here is a function that returns the neg-
 2 ative log-likelihood as a function of the three parameters.

```
> # The parameters are beta0, beta1, sigma
>
> neglogliketreg = function(pars, x, y, nu)
+ {
+   resid = y - (x*pars[2] + pars[1])
+   return((-1)*sum(dt(resid/pars[3], nu, log=TRUE)) +
+           length(x)*log(pars[3]))
+ }
```

- 3 Note how the `log=TRUE` argument to `dt()` is utilized to directly evaluate
 4 the log density.